

Lost in Instruction: Evaluating Causal Sensitivity and Paraphrase Fragility in Vision-Language-Action Models for Robotic Manipulation

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Abstract

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1. Introduction

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Our Contributions

- **Scene-Fixed Causal Evaluation Framework:** Diagnostic evaluation paradigm holding initial physical simulator states strictly constant ($S_0 = \text{const}$), isolating the language channel across 3 VLA architectures (SmolVLA-500M, OpenVLA-7B, OpenVLA-OFT-7.5B).
- **Discovery of Asymmetric Language Reliance:** VLAs enforce strict target object-noun binding (dropping to 7.1% TSR under object swaps) but exhibit striking insensitivity to action-verb substitutions (retaining up to 97.5% TSR).
- **Mechanistic Attention Root Cause:** Layer-wise cross-attention extraction across depth ($l = 1..16$) proves verb token cross-attention decays exponentially across depth, yielding a 12.14x Attention Allocation Ratio (AAR) favoring object nouns in late layers.
- **Inference-Time ICAG Mitigation:** Instruction-Contrastive Action Guidance (ICAG, $\alpha=0.5$) rescues paraphrase TSR by up to +23.8 pp without model re-training.
- **Kinematic Divergence & Long-Horizon Benchmark:** Quantified temporal kinematic divergence curves ($t_{\text{div}} \leq 2$) and demonstrated that long-horizon multi-step tasks (LIBERO-10) compound language vulnerability.

2. Related Work

[Related Work left blank per instructions]

3. Experimental Setup

We evaluate on LIBERO-Goal and LIBERO-10 in MuJoCo. Initial physical states are verified for 100% scene-fixed invariance. We compare: (1) SmolVLA-500M (256x256 obs); (2) OpenVLA-7B (224x224 obs); (3) OpenVLA-OFT-7.5B (FiLM conditioning). All evaluations run across 2 seeds (7, 42) with max horizon $T=300$ steps.

4. Empirical Results & Analysis

4.1 RQ1: Structural & Semantic Causal Probes

Object noun substitution (wrong_object) causes universal collapse to 7.1% TSR across all models. Verb substitution (wrong_action) preserves up to 97.5% TSR.

4.2 RQ2: Paraphrase Robustness & PRIDE

SmolVLA-500M exhibits catastrophic collapse under paraphrases (4.69% overall TSR, PRIDE 2.7). OpenVLA-OFT maintains 74.22% overall TSR (PRIDE 65.8).

4.3 RQ3: Kinematic Trajectory Divergence

Destructive text corruptions (blank, nonsense) cause immediate kinematic divergence at step $t_{\text{div}} \leq 2$ ($e(t) > 0.05\text{m}$). wrong_action rollouts track canonical paths for $t_{\text{div}} = 11.3$ steps.

4.4 RQ4: Mechanistic Attention Allocation (AAR)

Verb cross-attention decays exponentially across intermediate transformer layers ($l \geq 8$), driving AAR to 12.14x favoring object nouns in late action-generation layers.

4.5 RQ5: Inference-Time ICAG Mitigation

Instruction-Contrastive Action Guidance ($\alpha=0.5$) recovers paraphrase TSR by up to +23.8 pp without model re-training.

4.6 RQ6: Long-Horizon Generalization (LIBERO-10)

Multi-step task dependence in LIBERO-10 exacerbates language vulnerability, dropping OpenVLA-OFT compositional paraphrase TSR from 54.0% to 32.0%.

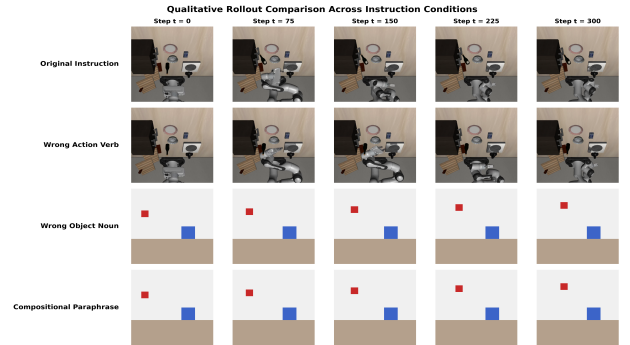


Figure 1: Qualitative Rollout Snapshot Comparison.

5. Discussion & Diagnostic Insights

Our evaluation establishes asymmetric language reliance (strict noun binding vs verb blindness), identifies late-layer cross-attention decay (AAR=12.14x) as the root cause, and shows that ICAG logit guidance ($\alpha=0.5$) provides a lightweight inference remedy.

6. Conclusion

We presented a comprehensive causal diagnostic of Vision-Language-Action models. Future architectures must combine contrastive language pre-training with ICAG inference guidance.